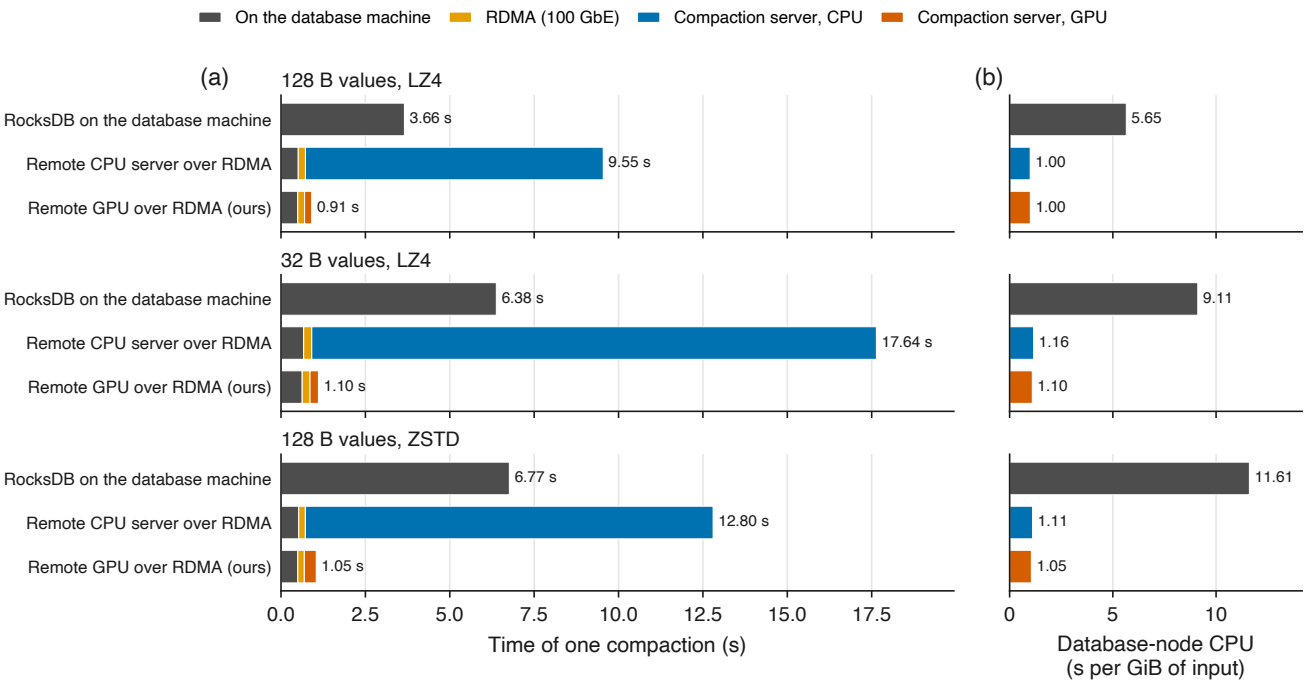
On the database machine
RDMA (100 GbE)
Compaction server, CPU
Compaction server, GPU
(a)
128 B values, LZ4
RocksDB on the database machine
Remote CPU server over RDMA
Remote GPU over RDMA (ours)
3.66 s
9.55 s
0.91 s
32 B values, LZ4
6.38 s
17.64 s
1.10 s
128 B values, ZSTD
6.77 s
12.80 s
1.05 s
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time of one compaction (s)
(b)
5.65
1.00
1.00
9.11
1.16
1.10
11.61
1.11
1.05
0
5
10
Database-node CPU
(s per GiB of input)